

Inflammation, Endothelial Dysfunction, and Frailty Over 3 Years in Survivors of COVID-19 ARDS vs Survivors of Pneumonia

Zain Khan, PhD; Ansley E. Jones, MHS; William M. Gardner, MPH; Claire F. McGroder, MD; Scarlett O. Murphy, MD; Christopher Depender, BS; Margarita Mira-Sanchez, MD; Charity O. Ogunlusi, MBBS, MPH; Christine K. Garcia, MD, PhD; Ying Wei, PhD; and Matthew R. Baldwin, MD

BACKGROUND: Inflammation and endothelial dysfunction are associated with increased severity and mortality in COVID-19 ARDS.

RESEARCH QUESTIONS: Do survivors of COVID-19 ARDS show greater inflammation, endothelial dysfunction, and frailty over 3 years compared with survivors of COVID-19 pneumonia?

STUDY DESIGN AND METHODS: We conducted a single-center prospective cohort study of 150 adults hospitalized during the first COVID-19 wave: 101 patients (67%) with ARDS and 49 patients (33%) with pneumonia requiring only nasal cannula oxygen. Participants were assessed at 4 months, 15 months, and 3 years for physical function, the frailty phenotype, and 9 circulating endothelial and inflammatory biomarkers. We tested associations of ARDS with biomarkers and physical function at each follow-up and via novel 3-year time-weighted average modeling of biomarker levels and functional measures by calculating the area under the curve divided by the time between follow-up. Analyses were adjusted for demographics and comorbidities and for multiple comparisons among biomarker classes.

RESULTS: In adjusted analyses, higher growth differentiation factor (GDF)-15 levels at hospital discharge and soluble intracellular adhesion molecule (sICAM)-1 levels were nominally and significantly associated with ARDS, respectively, and higher soluble tumor necrosis factor receptor (sTNFR)-1 levels at 3 years was associated nominally with ARDS. Of the 9 biomarkers, only higher GDF-15 levels were associated with ARDS in 3-year time-weighted analyses. Survivors of ARDS showed greater frailty at 4 months and in 3-year time-weighted average models. Of the 9 biomarkers, only GDF-15 at hospital discharge was associated with both ARDS and frailty.

INTERPRETATION: Our results show that survivors of COVID-19 ARDS exhibited higher levels of GDF-15 and sICAM-1 at discharge, as well as greater frailty, compared with survivors of COVID-19 pneumonia, but these distinctions largely converged by 15 months and 3 years of follow-up. Of all the biomarkers tested, only GDF-15 at hospital discharge was associated with subsequent frailty. These findings are preliminary and require validation, but suggest that targeting residual inflammation-related cellular stress at discharge could mitigate after COVID-19 ARDS frailty.

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KEY WORDS: ARDS; biomarkers; COVID-19; frailty; survivors

Take-Home Points

Study Question: Do survivors of COVID-19 ARDS have greater inflammation, endothelial dysfunction, and frailty over 3 years compared with survivors of COVID-19 pneumonia?

Results: Compared with survivors of COVID-19 pneumonia, survivors of COVID-19 ARDS have higher growth differentiation factor (GDF)-15 levels and greater frailty over 3 years—with these differences narrowing after 1 year—and higher GDF-15 level at hospital discharge is associated with increased frailty during the first year after hospitalization.

Interpretation: These preliminary findings support the novel hypothesis that GDF-15-related inflammation and cellular stress in survivors of COVID-19 ARDS is transient and mostly resolving within months, but may contribute to after COVID-19 ARDS frailty.

Greater severity of acute COVID-19 illness, especially COVID-19 ARDS requiring intensive care, is a major risk factor for persistent physical disability, postexertional malaise, and fatigue.^{1,2} However, the mechanisms underlying persistent disability and physical symptoms in survivors COVID-19 ICU admission are understood poorly. Higher levels of inflammation and endothelial dysfunction during acute COVID-19 are associated with greater severity of illness, prolonged mechanical ventilation, and higher short-term mortality,^{3,4} but whether inflammation and endothelial dysfunction persist in survivors of COVID-19 ARDS compared with survivors of COVID-19 pneumonia who required only nasal cannula oxygen therapy remains unknown. Higher levels of IL-6, IL-8, and soluble tumor necrosis factor receptor (sTNFR)-1 predict high inflammatory subphenotypes and higher mortality in non-COVID-19 and COVID-19 ARDS.^{3,5} In longitudinal studies of aging, higher levels of these inflammatory cytokines are implicated in the

development sarcopenia, frailty, physical disability, and symptoms of exhaustion,^{6,7} which mirror the physical limitations and symptoms observed in survivors of COVID-19 ARDS. Enhanced expression of IL-17 in the lung promotes downstream production of IL-6, IL-8, and sTNFR-1,^{8,9} and its inhibition is a common and successful strategy to reduce injury associated with inflammatory autoimmune diseases.¹⁰

Higher levels of circulating growth differentiation factor (GDF)-15, a biomarker released in response to hypoxemia, inflammation, oxidative stress, and mitochondrial dysfunction, are associated with greater multiorgan failure and higher mortality in non-COVID-19 and COVID-19 ARDS.^{11,12} We previously demonstrated in older adult survivors of acute respiratory failure that higher levels of circulating GDF-15 at hospital discharge were associated independently with muscle weakness and slower rates of recovery from physical disability.¹³

Biomarkers of endothelial dysfunction, including angiopoietin-2, soluble intercellular adhesion molecule (sICAM)-1, and soluble vascular cell adhesion molecule (sVCAM)-1 are associated with more severe and prolonged acute COVID-19.¹⁴ Survivors of COVID-19 ARDS with impaired gas exchange show elevated biomarkers of endothelial inflammation and dysfunction, including ICAM-1,¹⁵ and endothelial dysfunction has been implicated in the pathogenesis of physical frailty.¹⁶ Microthrombosis and hypercoagulable states involving von Willebrand factor-cleaving protease occur more frequently in COVID-19 ARDS than in non-COVID ARDS and are associated with greater disease severity and mortality.¹⁷⁻¹⁹

The primary scientific aim of this study was to test the hypothesis that survivors of COVID-19 ARDS experience greater inflammation, endothelial dysfunction, and frailty over 3 years after discharge compared with survivors of COVID-19 pneumonia. We assessed the aforementioned biomarkers to test our study hypothesis.

ABBREVIATIONS: GDF = growth differentiation factor; IQR = interquartile range; PASC = postacute sequelae of COVID-19; sICAM = soluble intracellular adhesion molecule; SPPB = Short Performance Physical Battery; sTNFR = soluble tumor necrosis receptor inhibitor; sVCAM = soluble vascular cell adhesion molecule; TWA = time-weighted average

AFFILIATIONS: From the Department of Biostatistics (Z. K. and Y. W.), Columbia University Mailman School of Public Health, and the Division of Pulmonary, Allergy, and Critical Care (A. E. J., W. M. G., C. F. M., S. O. M., C. D., M. M.-S., C. O. O., C. K. G., and M. R. B.),

Columbia University Vagelos College of Physicians & Surgeons, New York, NY.

CORRESPONDENCE TO: Matthew R. Baldwin, MD; email: mrb45@cumc.columbia.edu

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Study Design and Methods

Study Design and Participants

The Columbia University Irving Medical Center COVID-19 Acute Lung Injury/Acute Respiratory Distress Syndrome cohort has been described in detail previously.²⁰⁻²² We conducted a single-center prospective longitudinal cohort study of ambulatory, community-dwelling adults 21 years of age or older and hospitalized at the Milstein tertiary care and Allen community-level hospitals between March and May of 2020. We excluded those with a history of interstitial lung disease or lung transplantation. Sampling was weighted to include approximately 50% survivors of invasive mechanical ventilation at each follow-up.

We defined survivors from the COVID-19 Acute Lung Injury/Acute Respiratory Distress Syndrome cohort based on the global definition,²³ including those who required nonrebreather masks, high-flow nasal oxygen therapy, noninvasive positive-pressure ventilation, or invasive mechanical ventilation; those with bilateral patchy opacities not explained by pulmonary edema; and those with a PaO_2 to FiO_2 ratio of < 300 or SpO_2 to FiO_2 ratio of ≤ 315 if oxygenation saturation by pulse oximetry was $\leq 97\%$. Those who used nonrebreather facemasks were eligible for high-flow nasal oxygen or noninvasive positive-pressure ventilation, but no machines were available at the time because of the high numbers of patients with severe COVID-19. We defined survivors of COVID-19 pneumonia as those treated with only nasal cannula oxygen at 1-6 L/min during hospitalization ($n = 49$). We conducted 2 post hoc sensitivity analyses to test the robustness of our findings. First, we modeled COVID-19 acute lung injury severity ordinally as (1) nasal cannula oxygen only; (2) nonrebreather, high-flow nasal oxygen or noninvasive positive-pressure ventilation; or (3) invasive mechanical ventilation only. Second, we modeled COVID-19 acute lung injury severity as a dichotomous categorical variable of receiving invasive mechanical ventilation or not.

We enrolled 76 participants at the initial 4-month follow-up.²⁰ After the first year, additional funding enabled expansion of the cohort for subsequent follow-up.^{21,22} We invited the initial 4-month follow-up participants to participate at 15 months and 3 years. We enrolled 47 and 27 additional participants with the same inclusion criteria at 15 months and 3 years,

respectively, to increase and then maintain the sample size at about 100 participants. To minimize the potential for selection bias, we ensured that demographics, genomics, comorbidities, and acute COVID-19 hospitalization characteristics did not differ between those whom we did and did not follow up.^{21,22} The study was approved by the Columbia University Institutional Review Board (Identifier: AAAT5605 and AAAS7370).

Frailty and Functional Measurements

We measured the frailty phenotype at 4 months, 15 months, and 3 years. We assessed 4-m gait speed, grip strength, and exhaustion using the Cardiovascular Health Study methodology.²⁴ Low activity was measured with the Duke Activity Status Index,²⁵ which has strong construct validity in survivors of acute respiratory failure.²⁶ Weight loss was defined as > 10 pounds lost between acute COVID-19 hospital admission and the follow-up visit. Each frailty domain was assigned 1 point if present and 0 points if absent (range, 0-5). Patients were categorized as robust (0), prefrail (1-2), and frail (3-5), consistent with prior studies validating these phenotypes in community-dwelling older adults.²⁴ Additionally, we assessed the Short Performance Physical Battery (SPPB)²⁷ and 6-minute walk distance^{28,29} using established methods. In time-weighted average models, we categorized the SPPB score as ≤ 9 , the established cutoff that indicates physical limitation or frailty.³⁰

Biomarkers

Serum samples from the acute COVID-19 hospitalization were collected from residual daily morning clinical blood samples and cryopreserved at -80°C , usually within 4 hours. Available hospital discharge serum samples were collected within a median of 2 days (interquartile range [IQR], 4-1 days) before hospital discharge. Follow-up samples were collected prospectively via phlebotomy and centrifuged immediately, and serum was cryopreserved at -80°C . Biomarkers were quantified in duplicate and averaged using the Luminex xMAP bead-based multiplex assay platform (Miliopore Sigma). Biomarkers included IL-6, IL-8, IL-17a, TNFR-1, GDF-15, von Willebrand factor-cleaving protease, angiotensin-2, sICAM-1, and soluble sVCAM-1.

Statistical Analyses

First, we used generalized linear models to test associations of individual biomarkers at each follow-up

time (independent variables) with COVID-19 ARDS status (dependent variable). Then, we used generalized linear models to test associations of physical function measures at each follow-up time (independent variables) with prior COVID-19 ARDS status (dependent variable). We adjusted for age, sex, BMI, cardiovascular disease, COPD, asthma, diabetes, estimated glomerular filtration rate closest in time to hospital discharge,³¹ pack-years of smoking, and days from positive SARS-CoV-2 test results to follow-up. We did not control for acute hospitalization corticosteroid or IL-6 inhibitor receptor therapy or secondary sepsis because they represent potential mediators or effect modifiers, and not confounders. Nominal statistical significance was defined as $P < .05$ before correction for multiple comparisons. Statistical significance was defined as a P value remaining significant after Benjamini-Hochberg adjustment for multiple comparisons within biomarker classes (inflammation and endothelial function). Models were run in R version 4.4.1 software (R Foundation for Statistical Computing).

$$TWA = \frac{1}{t_n - t_1} \sum_{i=1}^{n-1} \frac{\ln(\text{Biomarker}(i)) + \ln(\text{Biomarker}(i+1))}{2} (t_{i+1} - t_i)$$

We sought to compare prospective repeated measures of biomarkers and physical function between survivors of COVID-19 ARDS and survivors of COVID-19 pneumonia across the 3-year follow-up.²⁰⁻²² At our moderate sample size, growth mixture models would be subject to nonconvergence, overparameterization, and poor class separation. The study design also precluded the use of a traditional longitudinal mixed-effects model, which would require specifying the biomarker as the longitudinal outcome with patient-specific random effects, and using the ARDS vs pneumonia group variable as the predictor, that is, reversing the dependent and independent variables in the longitudinal analysis.

To address our after ARDS study design, we developed a novel time-weighted average (TWA) modeling approach to evaluate longitudinal differences in biomarker and physical function levels between groups. First, because only 41 of 150 patients had biomarker data at hospital discharge and participated in all 3 follow-up visits, we used linear interpolation to impute missing values at

the 4-month and 15-month assessments. Missing values were estimated by drawing a straight line between the nearest observed time points, assuming a linear biomarker trajectory. This approach was motivated by 2 factors: biomarker values were natural log-transformed before interpolation, which linearizes exponential decay patterns, and our moderate sample size limited feasibility of fitting more complex nonlinear trajectory models. Biomarker data were reshaped into a long format indexed by patient and follow-up visit time. Individual trajectories were summarized using TWA to account for irregular observation intervals (0-4 months, 4-15 months, and 15-36 months). TWA was computed as the area under the linearly interpolated curve divided by the total follow-up time:

$$TWA = \frac{\int_{t_1}^{t_n} \ln(\text{Biomarker}(t)) dt}{t_n - t_1} = \frac{AUC}{t_n - t_1}$$

The discrete analog of the TWA, applied in practice, used the trapezoidal rule to estimate the area under the curve (AUC).

For repeated categorical measures (frailty phenotype category and weak grip strength), where interpolation was not applicable, we calculated the time-weighted mode (category in which the participant spent the greatest proportion of observed time). This approach assigned each observed category to persist until the next measurement, calculating the total duration spent in each category, and selecting the category with the largest cumulative duration as the representative value. Analyses were performed in Python (Python Software Foundation) and R.

We used ordinal logistic and generalized linear models to test post hoc longitudinal associations of GDF-15 and sICAM-1 at hospital discharge with subsequent frailty category and SPPB score, because these biomarkers and physical function measures were found to be associated with prior COVID-19 ARDS. We adjusted for the same covariables described for the generalized linear models.

Results

Participant Characteristics

Among all 150 study participants, age, sex, race and ethnicity, and proportions of participants with cardiometabolic and pulmonary comorbidities remained similar or identical across study visits, despite attrition and resampling (Table 1, e-Fig 1). Two participants from the 4-month follow-up died during

the study period. One declined to participate at the 15-month follow-up because of hospice admission for terminal abdominal infection, and another died of acute leukemia between the 15-month and 3-year follow-up. Fewer participants at 4 months had received invasive mechanical ventilation than at 15 months or 3 years (42% vs 51% and 49%, respectively), but median admission Sequential Organ Failure Assessment scores were similar across visits.

TABLE 1] Patient Characteristics by Follow-Up Period

Characteristic	All (N = 150)	4-Mo Follow-up (n = 76)	15-Mo Follow-up (n = 104)	3-Y Follow-up (n = 102)
Age, y	56 (13)	54 (14)	54 (11)	56 (12)
Male sex	83 (56%)	45 (61%)	61 (59%)	55 (54%)
BMI, kg/m ²	33 (8.1)	32 (6.9)	32 (6.8)	32 (6.2)
Hispanic ethnicity	100 (67%)	43 (57%)	62 (60%)	69 (68%)
Race				
White	55 (37%)	30 (39%)	30 (29%)	40 (40%)
Black	35 (23%)	22 (30%)	28 (27%)	25 (25%)
Asian	5 (3.3%)	1 (1%)	4 (4%)	4 (4%)
Other	54 (36%)	23 (30%)	42 (40%)	33 (32%)
Comorbidities				
Ever smoking ^a	58 (39%)	31 (41%)	40 (38%)	39 (38%)
Active smoking	4 (2.7%)	2 (3%)	3 (3%)	2 (2%)
COPD	7 (4.7%)	4 (5%)	4 (4%)	2 (2%)
Asthma	30 (20%)	18 (27%)	24 (23%)	22 (22%)
Cardiovascular disease	19 (13%)	7 (9%)	13 (13%)	13 (13%)
eGFR	95 (45)	96 (44)	98 (45)	98 (45)
COVID-19 illness characteristics				
Admission SOFA score	3 (2-5)	3.5 (3-6)	3 (2-5)	3 (2-5)
Steroid use	67 (47%)	39 (51%)	43 (41%)	47 (46%)
IL-6 receptor inhibitor use	31 (21%)	17 (22%)	26 (25%)	23 (23%)
Oxygen use				
None	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Nasal cannula	49 (32%)	23 (30%)	31 (30%)	32 (31%)
Nonrebreather	28 (19%)	17 (22%)	18 (17%)	18 (18%)
NIPPV or HFNO	4 (2.7%)	4 (5%)	2 (2%)	2 (2.0%)
Mechanical ventilation	68 (46%)	32 (42%)	53 (51%)	50 (49%)
Ventilator days ^b	34 (14-46)	31 (12-42)	35 (18-46)	35 (17-46)
Hospitalization duration, d	18 (7-42)	18 (7-35)	23 (8-47)	23 (7-46)
Discharge disposition				
Home	101 (67%)	54 (71%)	67 (64%)	65 (64%)
Acute rehabilitation	40 (27%)	19 (25%)	33 (32%)	28 (27%)
Subacute rehabilitation or skilled care	9 (6%)	3 (4%)	4 (4%)	9 (9%)

Data are presented as No. (%), mean (SD), or median (interquartile range). eGFR = estimated glomerular filtration rate (based on CKD-EPI equation); HFNO = high-flow nasal oxygen; NIPPV = noninvasive positive pressure ventilation; SOFA = Sequential Organ Failure Assessment.

^aFormer or active smoking.

^bAmong those who received invasive mechanical ventilation.

TABLE 2] Patient Characteristics by COVID-19 Severity Status: Pneumonia vs ARDS

Characteristic	Pneumonia (n = 49)	ARDS (n = 101)	P Value
Demographics			
Age, y	59 (12)	54 (13)	.027
BMI, kg/m ²	32 (7.8)	32 (7.5)	.691
Sex			
Female	29 (59%)	37 (37%)	
Male	20 (41%)	64 (63%)	.015
Ethnicity			
Hispanic or Latino	27 (55%)	73 (73%)	.046
Race			
White	19 (39%)	36 (36%)	
Black	17 (35%)	18 (18%)	
Asian	1 (2.0%)	4 (4.0%)	
Other	2 (4.0%)	2 (2.0%)	
Multiple or unknown	10 (20%)	40 (40%)	.071
Comorbidities			
Ever smoking	25 (51%)	33 (33%)	.047
COPD	1 (2.0%)	6 (5.9%)	.516
Asthma	11 (22%)	19 (19%)	.761
Cardiovascular disease	7 (14%)	12 (12%)	.878
Diabetes	12 (5%)	29 (29%)	.727
COVID-19 characteristics			
Admission SOFA score			
IL-6 receptor inhibitor use	0 (0.0%)	31 (31%)	< .001
Corticosteroids use	4 (11%)	49 (49%)	< .001
Hospital LOS, d	5 (3-7)	33 (18-50)	< .001
Ventilation duration, d	N/A	34 (14-46)	< .001
Discharge disposition			< .001
Home	48 (98%)	53 (52%)	
Acute rehabilitation	0 (0%)	40 (40%)	
Subacute rehabilitation or skilled care	1 (2.0%)	8 (8%)	

Data are presented as No. (%), mean (SD), or median (interquartile range) unless otherwise indicated. LOS = length of stay; N/A = not applicable; SOFA = Sequential Organ Failure Assessment.

Of all participants, 49 participants (33%) had COVID-19 pneumonia and 101 participants (67%) met study criteria for COVID-19 ARDS (Table 2).

Compared with patients with pneumonia, those with ARDS were younger (mean [SD] age, 54 [13] years vs 59 [12] years; $P = .027$), more often male (63% vs 41%; $P = .015$), and predominantly Hispanic (73% vs 55%; $P = .046$). Comorbidity prevalence was similar between groups. Patients with ARDS experienced longer median hospital stays (33 days [IQR, 18-50 days] vs 5 days [IQR, 3-7 days]; $P < .001$).

Among those requiring invasive mechanical ventilation,

the median duration of ventilation was 34 days (IQR, 14-46 days).

Biomarkers in Generalized Linear Model Analyses

Substantial overlap was found in biomarker levels over the 3-year study period between participants with COVID-19 pneumonia and participants with COVID-19 ARDS, with some exceptions (Fig 1). Mean GDF-15 and sICAM-1 levels seemed higher in participants with ARDS at hospital discharge, but converged with levels in participants with pneumonia by 4 months. Mean IL-8 levels seemed higher in

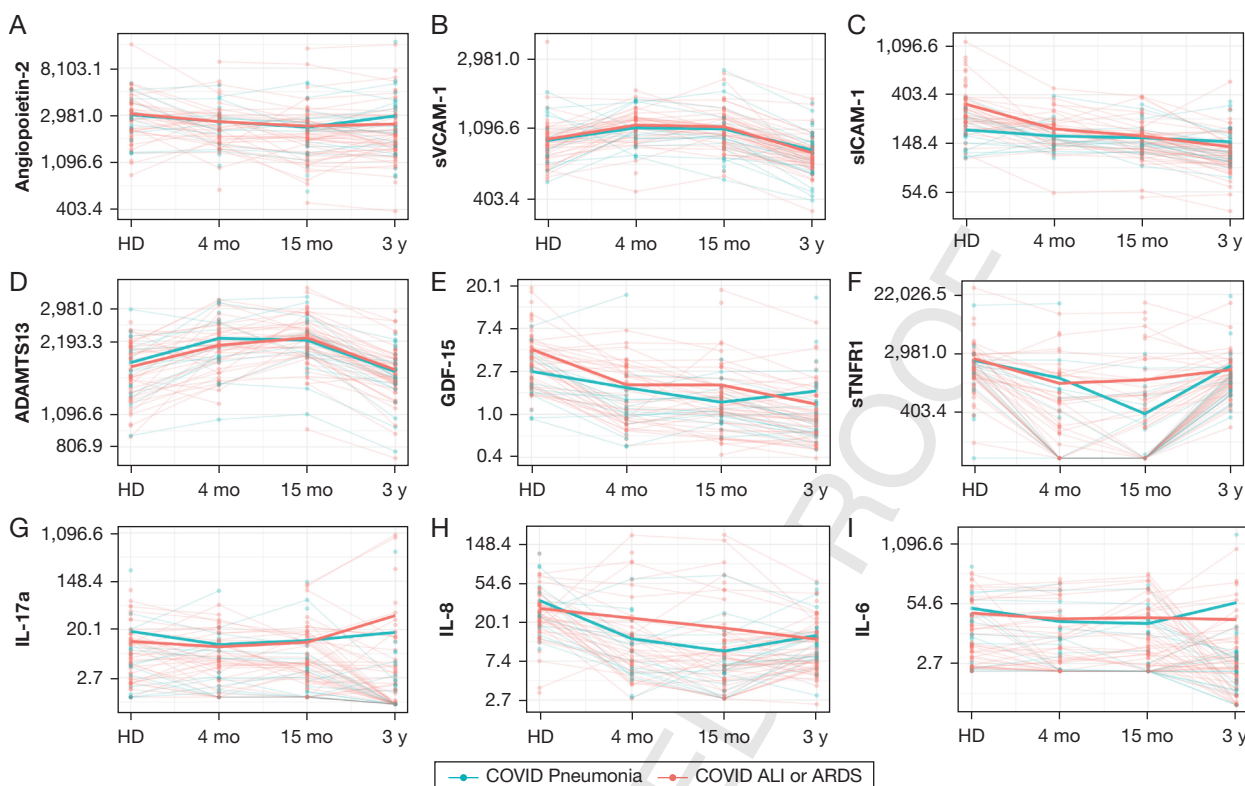


Figure 1 – A-I, Graphs showing longitudinal trajectories of biomarkers stratified by prior COVID-19 ARDS vs COVID-19 pneumonia: angiopoietin-2 (A), sICAM-1 (B), sVCAM-1 (C), ADAMTS13 (D), GDF-15 (E), sTNFR1 (F), IL-17a (G), IL-8 (H), and IL-6 (I). Given the right-skew of biomarker data, the y-axis is a natural log scale. Mean biomarker values at a particular time point appear in boldface. ADAMTS13 = von Willebrand factor-cleaving protease; ALI = acute lung injury; GDF = growth differentiation factor; sICAM = soluble intracellular adhesion molecule; sTNFR-1 = soluble tumor necrosis receptor inhibitor-1; sVCAM = soluble vascular cell adhesion molecule.

survivors of ARDS at 4 and 15 months, and mean sTNRI levels seemed higher at 15 months.

In adjusted models, higher levels of GDF-15 at hospital discharge and sTNFR-1 at 3 years nominally were associated significantly with ARDS. For every natural log-fold increase in GDF-15, there was 4.5 times the odds of having had COVID-19 ARDS (95% CI, 1.2-21). For every natural log-fold increase in sTNFR-1, there was 3.6 times the odds of having had COVID-19 ARDS (95% CI, 1.2-14). However, these associations did not remain significant after correction for multiple comparisons. Only sICAM-1 at hospital discharge remained associated significantly with ARDS after multiple comparison adjustment (OR, 27 [95% CI, 3.3-443]) (Fig 2). All other biomarker associations at the 3 follow-up time points were not significant. In sensitivity analyses, GDF-15 and sICAM-1 levels at discharge remained highly associated significantly with greater COVID-19 acute lung injury severity and use of invasive mechanical ventilation, but sTNFR-1 level at 3-years was not (e-Table 1).

Physical Function in Generalized Linear Model Analyses

At 4 months, participants with COVID-19 ARDS showed higher frailty phenotype scores than participants with pneumonia (median, 1 [IQR, 1-3] vs 0 [IQR, 0-1]) (Fig 3A-3D, e-Table 2). Only participants with ARDS showed frailty scores of 4 or 5, and weak grip strength was nearly twice as common (62% vs 30%). By 15 months and 3 years, frailty scores and weak grip prevalence were similar between survivors of COVID-19 ARDS and survivors of COVID-19 pneumonia.

Six-minute walk distance was similarly low at 4 months for participants with ARDS and participants with pneumonia (median, 70% predicted [IQR, 56%-79% predicted] vs 71% predicted [IQR, 55%-78% predicted]), but improved more in survivors with pneumonia at 15-month and 3-year follow-ups. SPPB was assessed only at 15 months and 3 years, with similar mean scores between groups. In adjusted models, higher frailty category and weak grip at 4 months and

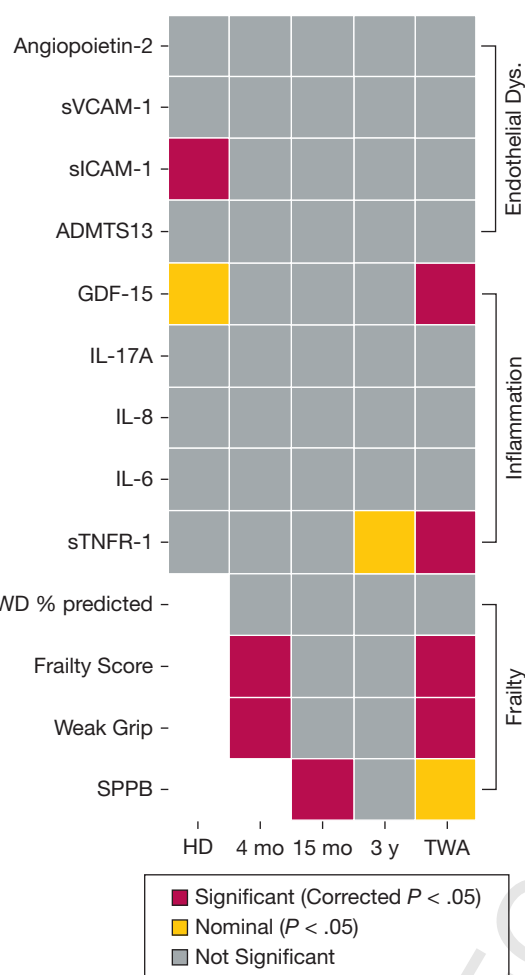


Figure 2 – Heatmaps showing the statistical significance of adjusted associations of biomarkers and physical function with prior COVID-19 ARDS vs COVID-19 pneumonia. Associations are at each time point using generalized linear models and using time-weighted average models. Adjusted model covariables are age, sex, BMI, cardiovascular disease, COPD, asthma, diabetes, estimated glomerular filtration rate closest in time to hospital discharge, pack-years of smoking, and days from positive SARS-CoV-2 test results to follow-up. The Benjamini-Hochberg method was used to assess significance after adjusting for multiple comparisons. 6MWD = 6-minute walking distance; ADAMTS13 = von Willebrand factor-cleaving protease; ALI = acute lung injury; GDF = growth differentiation factor; SPPB = Short Performance Physical Battery; sICAM = soluble intracellular adhesion molecule; sTNFR-1 = soluble tumor necrosis receptor inhibitor-1; sVCAM = soluble vascular cell adhesion molecule; TWA = time-weighted average.

lower SPPB score at 15 months each were associated independently with prior ARDS (Fig 2, e-Table 3).

Biomarkers and Physical Function in Time-Weighted Average Modelling Analyses

Unadjusted TWA biomarker z scores were visualized using ridge plots (Fig 4). GDF-15 and sICAM-1 distributions were slightly higher in survivors of COVID-19 ARDS vs survivors of COVID-19 pneumonia, likely reflecting the observed elevated levels

at hospital discharge. In adjusted TWA models, higher GDF-15 (OR, 14 [95% CI, 3-92]) and sTNFR-1 (OR, 2.1 [95% CI, 1.2-3.9]) levels each were associated independently with prior ARDS (Fig 2).

In adjusted TWA models, higher frailty score, weak grip, and lower SPPB scores were associated significantly with prior COVID-19 ARDS. Lower 6-minute walk distance % predicted was not associated significantly with prior COVID-19 ARDS (Fig 2, e-Table 3).

Post Hoc Biomarker and Physical Function Analyses

In adjusted analyses, GDF-15 (OR, 3.0 [95% CI, 1.2-8.4]), but not sICAM-1 (OR, 1.2 [95% CI, 0.4-3.8]), was associated significantly with higher frailty category at 4 months (e-Table 4). For every natural log-fold increase in GDF-15 at hospital discharge, there was 3.0 times the odds of being in a higher frailty category (95% CI, 1.2-8.4). GDF-15 (OR, 2.5 [95% CI, 0.8-9.5]) and sICAM-1 (OR, 1.7 [95% CI, 0.5-6.5]) were not associated significantly with the time-weighted average frailty category. Neither biomarker at hospital discharge was associated with SPPB score at 15 months or via the time-weighted average calculation.

Discussion

In one of the largest North American cohorts of survivors of severe and critical COVID-19, we found that survivors of COVID-19 ARDS exhibited higher levels of GDF-15 and sICAM-1, as well as greater frailty, compared with survivors of COVID-19 pneumonia, but these distinctions largely converged by 15 months and 3 years of follow-up. Despite no consistent biomarker differences detected after 1 year, survivors of ARDS showed greater GDF-15 levels and frailty in time-weighted models over 3 years, and GDF-15 level at hospital discharge was associated with frailty at 4 months. Although these findings are preliminary and require validation, they support the novel hypothesis that GDF-15-related inflammation and cellular stress in survivors of ARDS is transient, mostly resolving within months, but may contribute to more prolonged frailty.

Studies before the COVID-19 pandemic suggest that inflammation and endothelial dysfunction persist in some survivors of respiratory failure and sepsis. Greater inflammation at hospital discharge is associated with 1-year mortality in survivors of pneumonia,³² and inflammation at 3 months after ICU discharge is associated with worse mobility.³³ Higher levels of GDF-15 at hospital discharge are associated with weakness

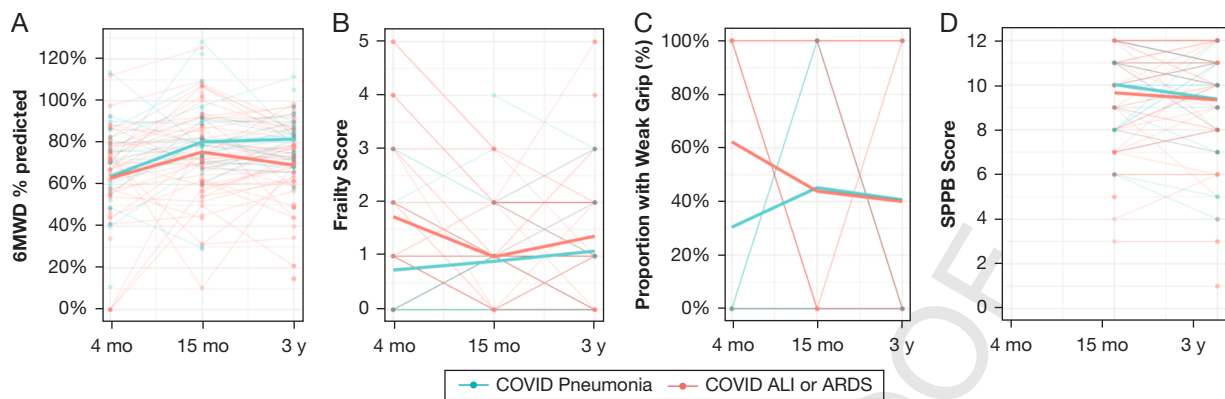


Figure 3 – A-C, Graphs showing longitudinal trajectories of physical function stratified by prior COVID-19 ARDS vs COVID-19 pneumonia: 6-minute walk distance % predicted (A), grip strength (B), and frailty phenotype score (C). Boldface lines connect mean physical function values at each follow-up time. 6MWD = 6-minute walk distance; ALI = acute lung injury; SPPB = Short Performance Physical Battery.

and slower functional recovery in older adult survivors of respiratory failure.¹³ Survivors of sepsis have a high risk of cardiovascular events, suggestive of underlying endothelial dysfunction.³⁴ In a United Kingdom-based study of survivors of COVID-19, the subgroup with the most severe physical, mental, and cognitive impairments at 6 months showed the highest C-reactive protein levels and the highest proportion of survivors of COVID-19 ARDS.³⁵

Our findings of elevated sICAM-1 and GDF-15 at hospital discharge are consistent with these other studies that suggest that persistent inflammation and endothelial dysfunction in the first few months after critical illness may drive worse outcomes. We add to the after respiratory failure literature by showing that these

elevations normalize to that of survivors of COVID-19 pneumonia treated with nasal cannula within the first year. We also observed a nominal association between prior COVID-19 ARDS and higher levels of sTNFR-1 at the 3-year follow-up that likely contributed to the observation of an association of higher levels of sTNFR-1 with prior ARDS in the 3-year TWA model. Although this finding is consistent with our study hypothesis, it could be spurious because it seems biologically implausible that ARDS would lead to higher sTNFR-1 levels only 3 years later. Additionally, the association did not remain significant in sensitivity analyses.

At least one-half of survivors of COVID-19 ARDS and non-COVID-19 ARDS experience persistent muscle weakness, leading to greater disability, health care costs,

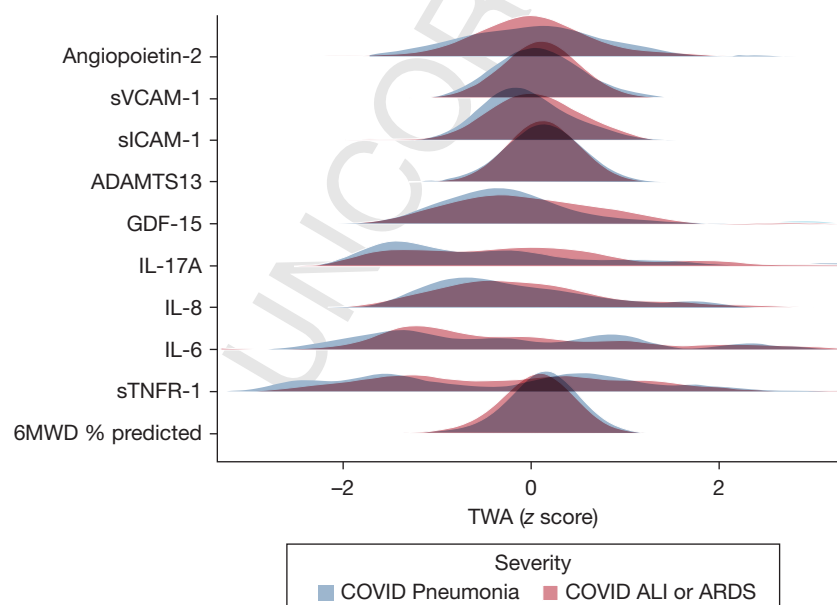


Figure 4 – Ridge plot showing the distribution of TWA serum biomarkers levels by prior COVID-19 ARDS vs COVID-19 pneumonia. The x-axis shows the standardized (z score) TWA values of each of the biomarkers and 6MWD % predicted. 6MWD = 6-minute walk distance; ADAMTS13 = von Willebrand factor-cleaving protease; ALI = acute lung injury; GDF = growth differentiation factor; sICAM = soluble intracellular adhesion molecule; sTNFR = soluble tumor necrosis receptor inhibitor; sVCAM = soluble vascular cell adhesion molecule; TWA = time-weighted average.

and mortality.^{2,36-43} No targeted therapies exist to improve muscle strength and achieve recovery because underlying mechanisms remain unclear. Our time-weighted average modeling approach advances long-term critical illness outcomes research by enabling the integration of repeated biomarker and clinical measurements with prior acute illness factors, even at modest sample sizes. The model addresses key analytic challenges: variable intervals between follow-up visits, attrition and missed visits (via interpolation), and high-dimensional data (via area under the curve reduction), thereby increasing statistical power. Using this method, we identified an association of GDF-15 with both ARDS and subsequent frailty. This supports a speculative hypothesis that GDF-15 may mediate and serve as a modifiable target for after COVID-19 ARDS-related frailty. Notably, in an early-phase trial of patients with cancer cachexia and elevated GDF-15 levels, inhibition of GDF-15 with the humanized monoclonal antibody ponesegromab increased weight gain, lumbar skeletal muscle index, and physical activity.⁴⁴

Our study has limitations. We studied a cohort from a single center enrolled during the first wave of the COVID-19 pandemic, and the study lacks external validation. Our results should be considered preliminary and hypothesis generating. Hospital discharge biomarkers were assessed from residual daily morning clinical blood samples, and study patients likely showed varying clinical statuses at hospital discharge. These factors may contribute to variability in the biomarker levels observed at hospital discharge. After the 4-month visit, we enrolled additional participants to increase and then maintain the study sample size at about 100 participants for the 15-month and 3-year visits. Demographics, genomics, comorbidities, and acute COVID-19 hospitalization characteristics did not differ substantially between those who did and did not undergo follow-up and did not change significantly across follow-up visits,^{21,22} but it remains possible that unmeasured selection bias was introduced. The survivors of ARDS on average were 5 years younger than survivors of COVID-19 pneumonia, which could indicate a potential healthy survivor selection bias of survivors of ARDS that, if present, likely would bias our observed associations towards the null. Because this was an inception cohort, we did not assess frailty before acute COVID-19 hospitalization, and thus cannot distinguish between preexisting and

incident frailty. However, we suspect that most frailty was incident, given that participants were ambulatory community-dwelling adults with a relatively young mean (SD) age of 55 (11) years. We used interpolation for missing longitudinal biomarker data so that imputed values remained consistent with an individual's biomarker trajectory. Linear interpolation assumes that biomarker changes occur at a constant rate between measurements, which may not reflect the underlying biological trajectory; however, this approach preserves individual-level trends. At our moderate sample size, it is difficult to estimate within-patient correlation over time using multiple imputation. We selected 9 biomarkers a priori based on their roles in the pathogenesis and severity of acute COVID-19 and frailty.^{3,4,6,7,16} Future studies should investigate novel biomarkers using other multiplex biomarker assays and plasma proteomics.

Interpretation

When compared with patients with COVID-19 pneumonia requiring only nasal cannula oxygen, survivors of COVID-19 ARDS have higher levels of GDF-15 at hospital discharge and frailty at 4 months that contributes to higher levels of GDF-15 and frailty averaged across 3 years. Although these results are preliminary and require validation, they suggest that GDF-15-related inflammation and cellular stress are present at hospital discharge, resolve within months, yet remain associated with frailty that is greater, on average, over 3 years. Our time-weighted average modeling approach may facilitate external validation of these findings in other cohorts of survivors of COVID-19 and critical illness.

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